

BitLocker GPO Configuration

The goal of this configuration is to prepare domain computers to use BitLocker in a controlled way.

In this setup, BitLocker is enabled manually on each computer. After that, a scheduled task deployed by GPO runs a PowerShell script that backs up the BitLocker recovery key to Active Directory.

The configuration is divided into two parts:

BitLocker policies

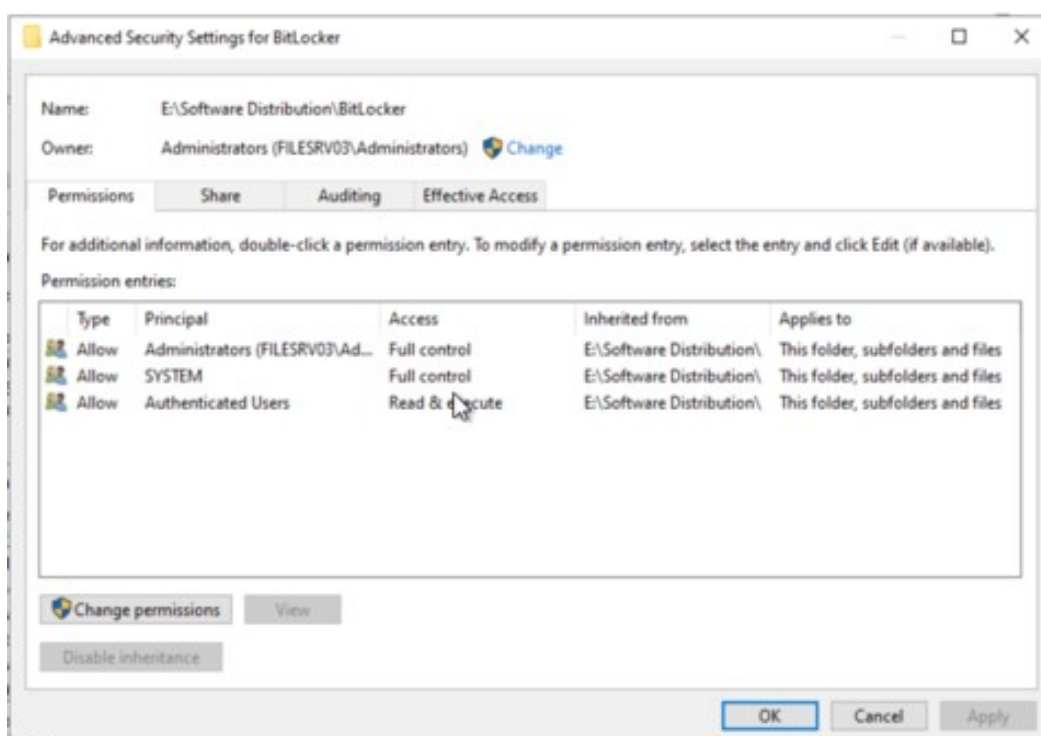
Scheduled task to run the backup script

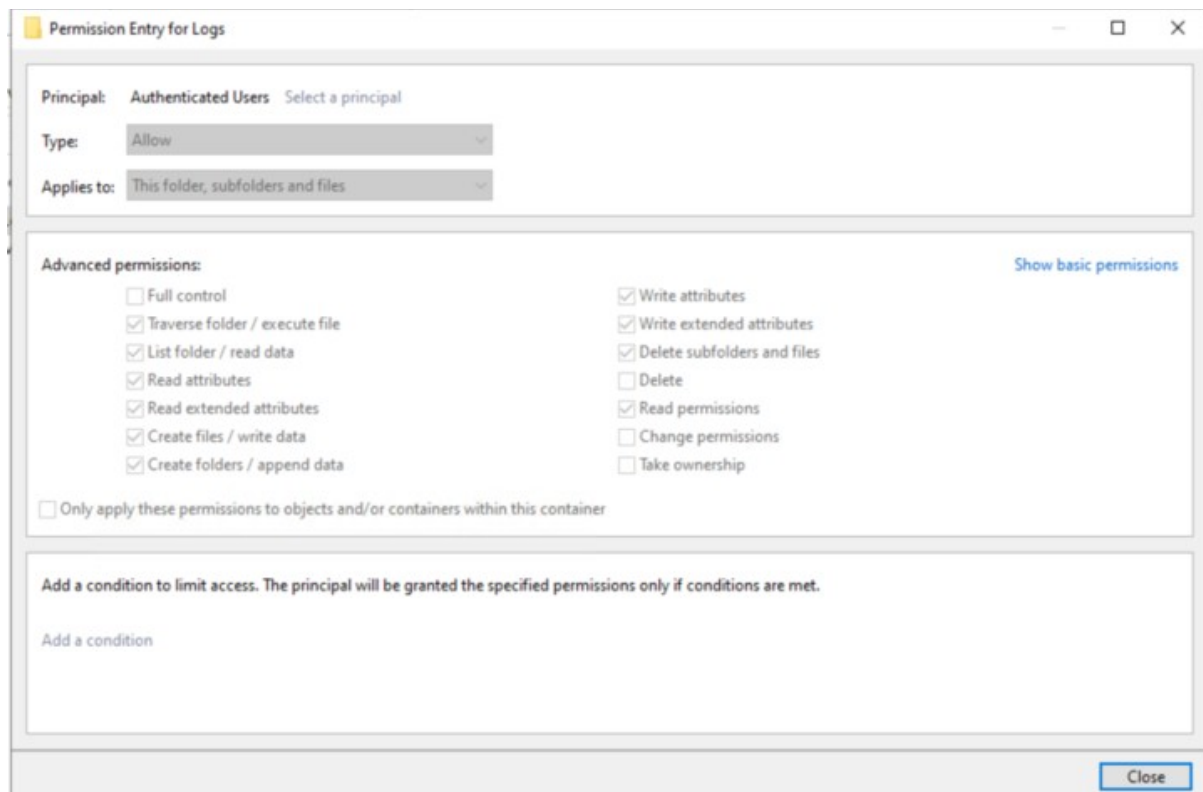
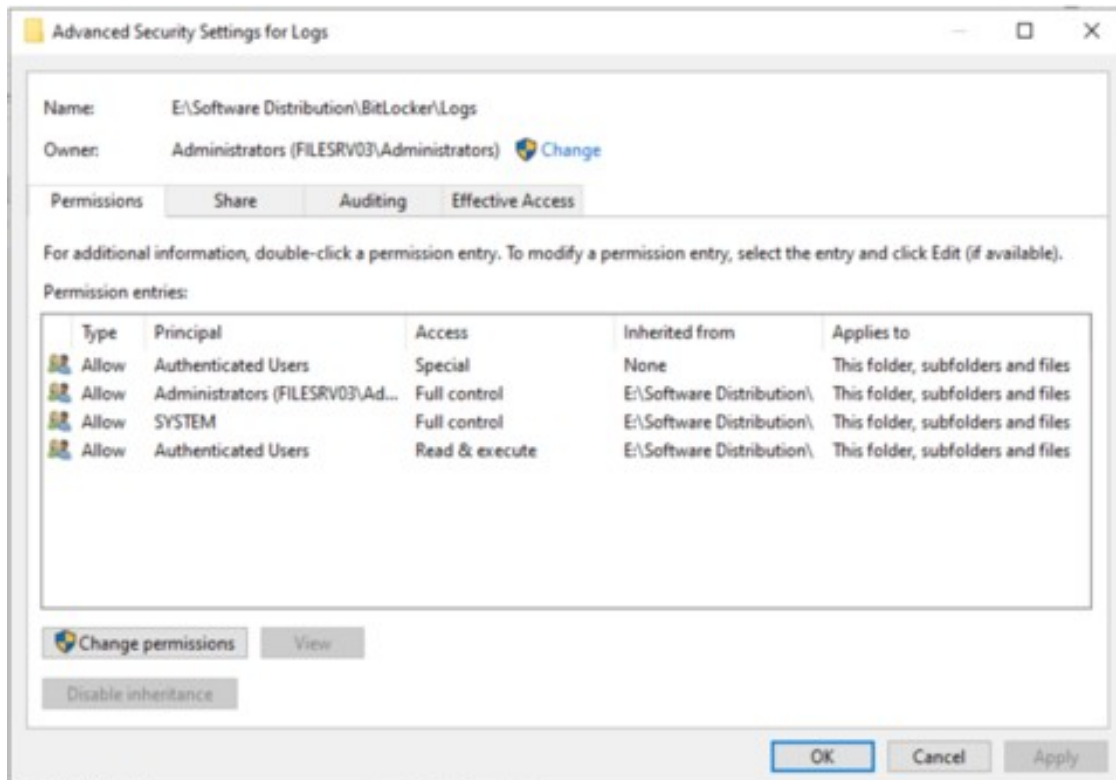
This is the folder path:

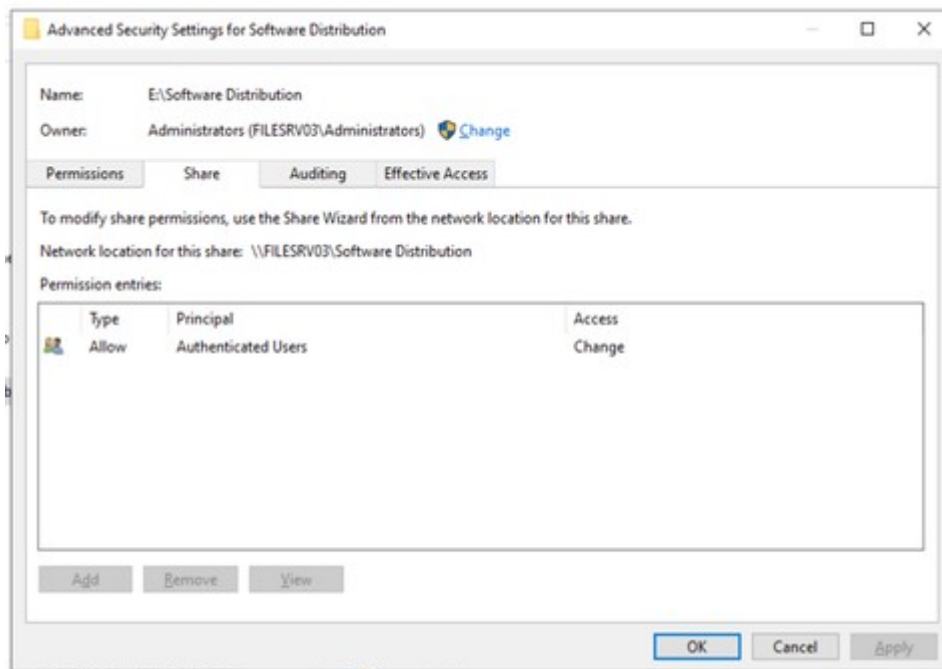
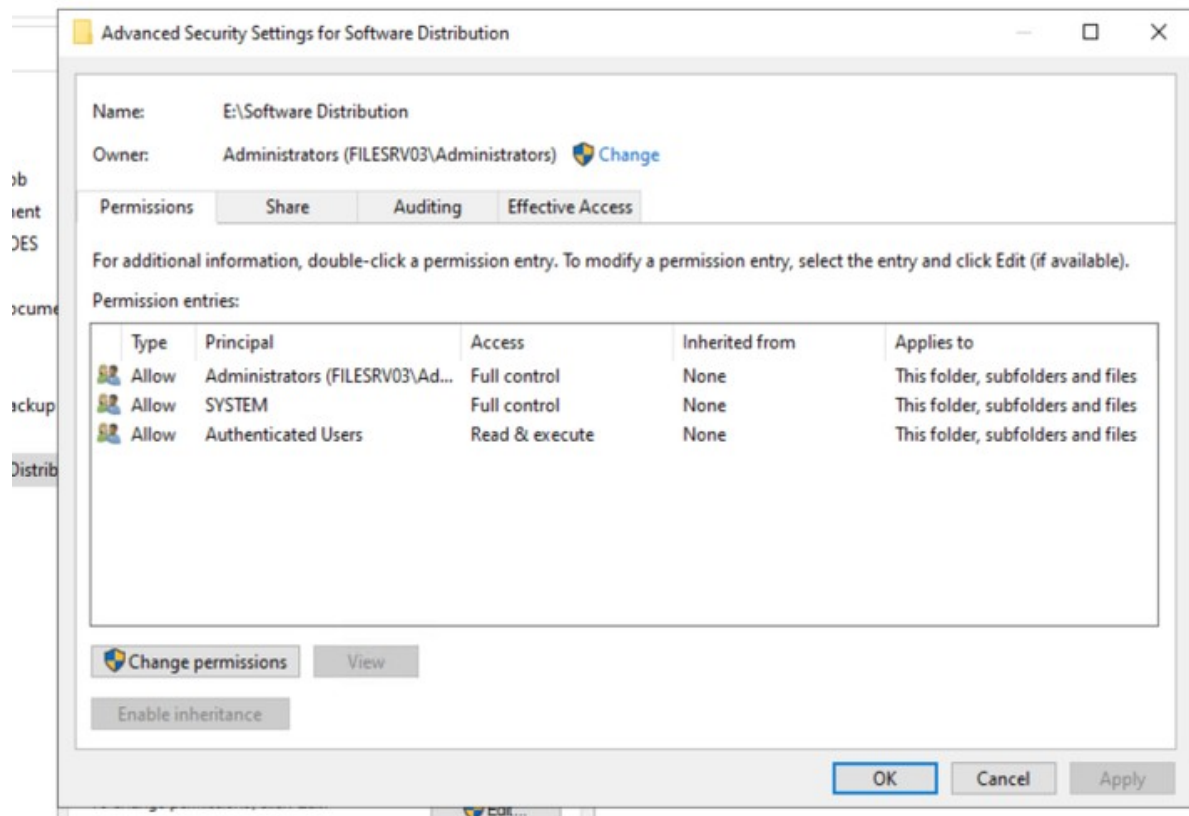
E:\Software Distribution\BitLocker\Manually-Backup-BitLocker-RecoveryKey-To-AD.ps1

E:\Software Distribution\BitLocker\

E:\Software Distribution\BitLocker\Logs







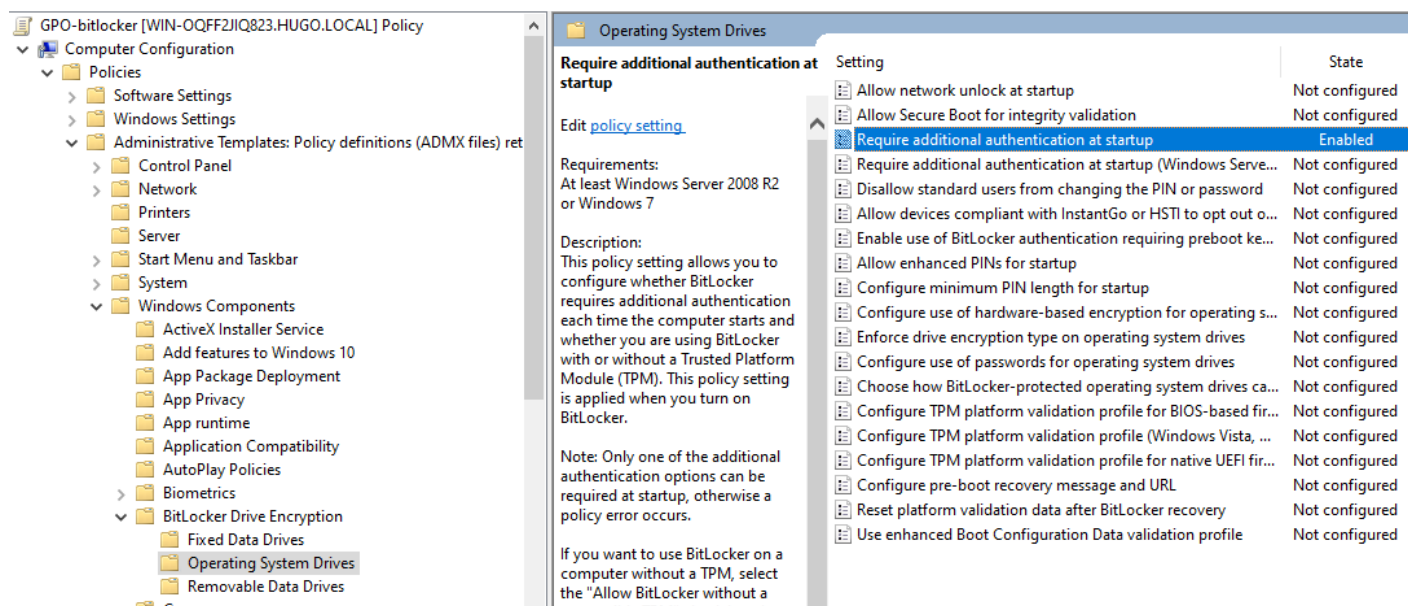
1. BitLocker startup authentication policy

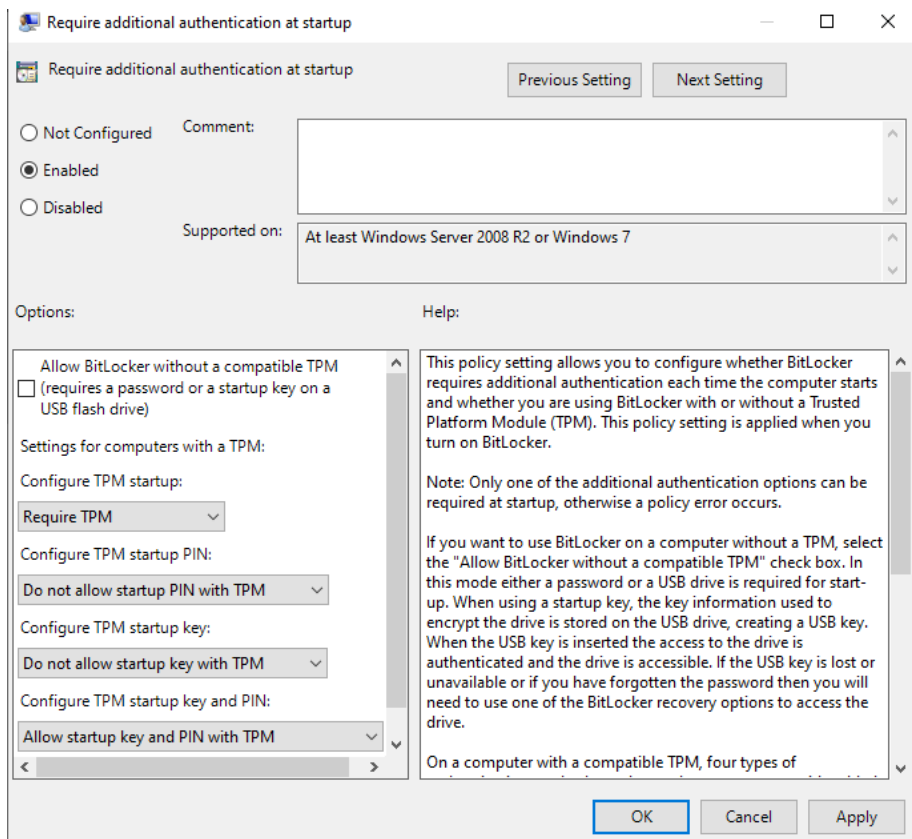
The first policy configures how BitLocker behaves when the computer starts.

The goal is to use the computer's TPM so the user does not need to enter a PIN or connect a USB key at startup.

Policy path:

Computer Configuration → Policies → Administrative Templates → Windows Components → BitLocker Drive Encryption → Operating System Drives → Require additional authentication at startup





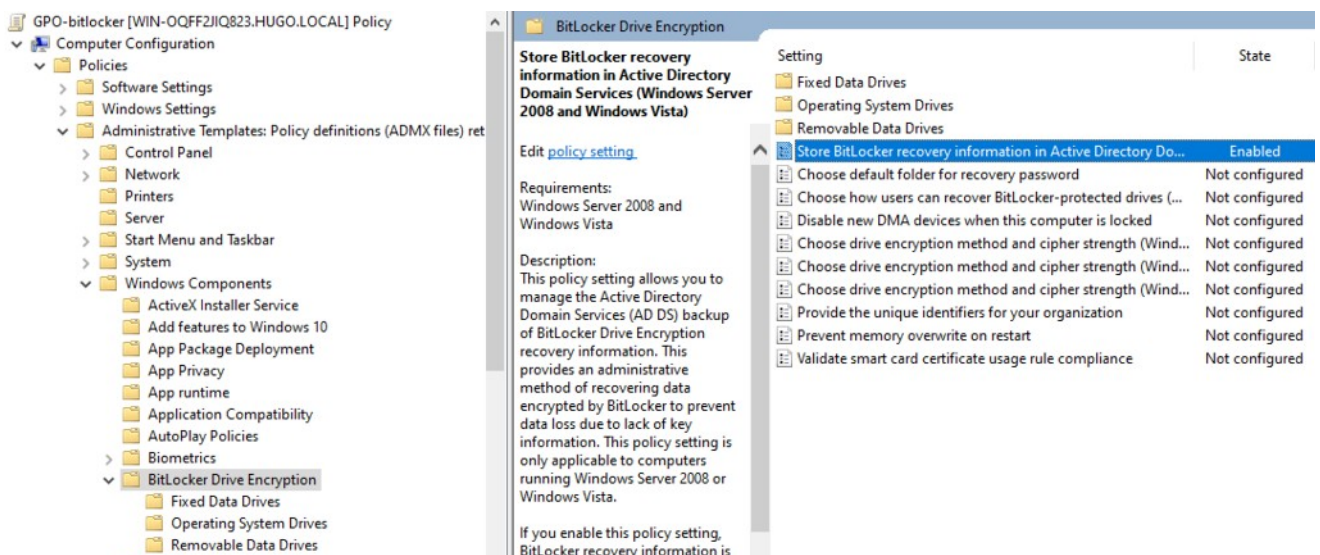
2. Store BitLocker recovery information in Active Directory

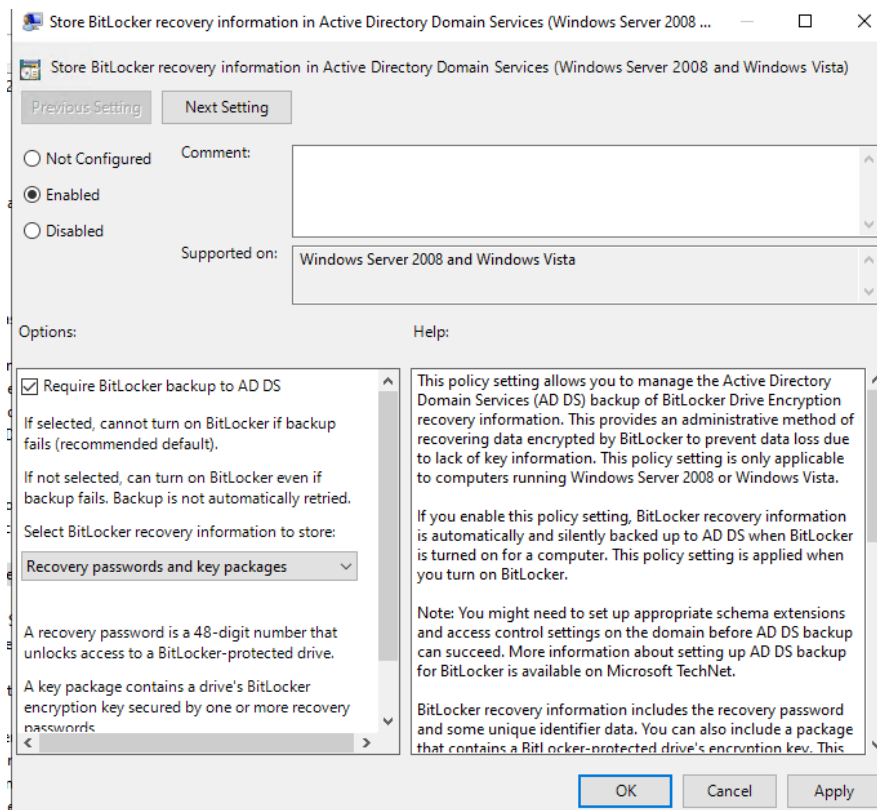
The next policy allows BitLocker recovery information to be stored in Active Directory.

This is important because, if a computer asks for the BitLocker recovery key, the administrator can recover it from Active Directory instead of relying on a local copy.

Policy path:

Computer Configuration → Policies → Administrative Templates → Windows Components → BitLocker Drive Encryption → Store BitLocker recovery information in Active Directory Domain Services





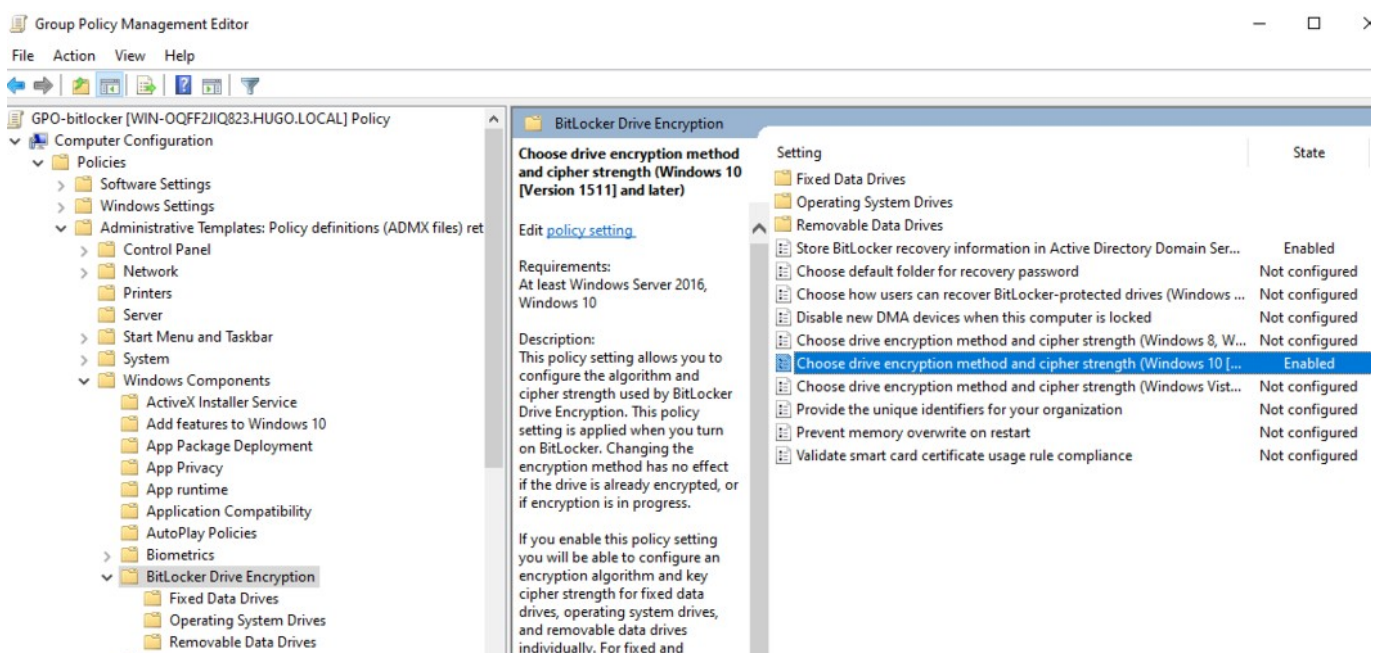
3. Drive encryption method and cipher strength

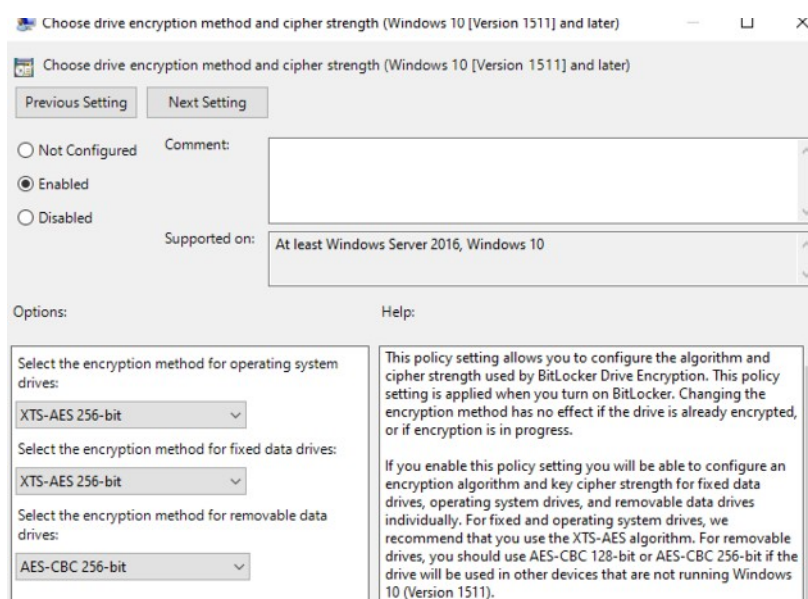
This policy defines the encryption method used by BitLocker.

In this case, the configuration uses AES 256-bit encryption for stronger protection.

Policy path:

Computer Configuration → Policies → Administrative Templates → Windows Components → BitLocker Drive Encryption → Choose drive encryption method and cipher strength





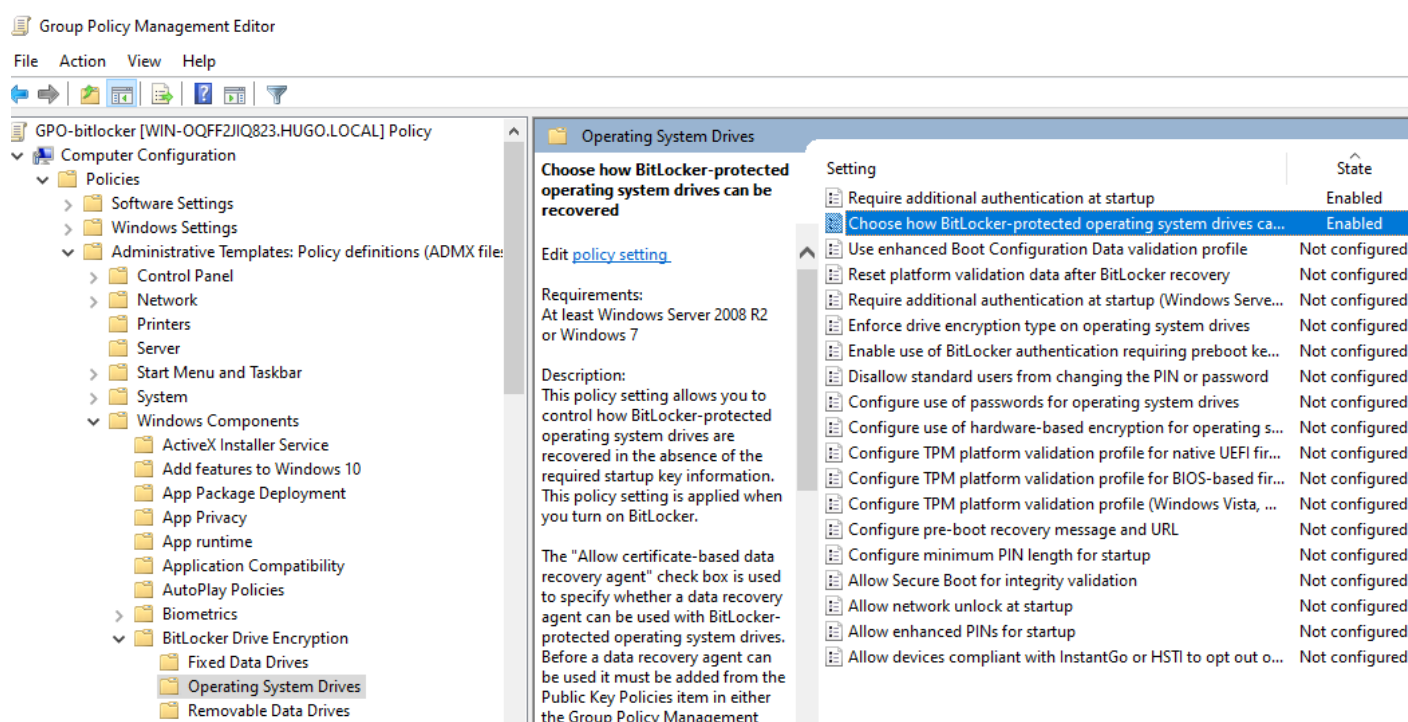
4. Operating system drive recovery policy

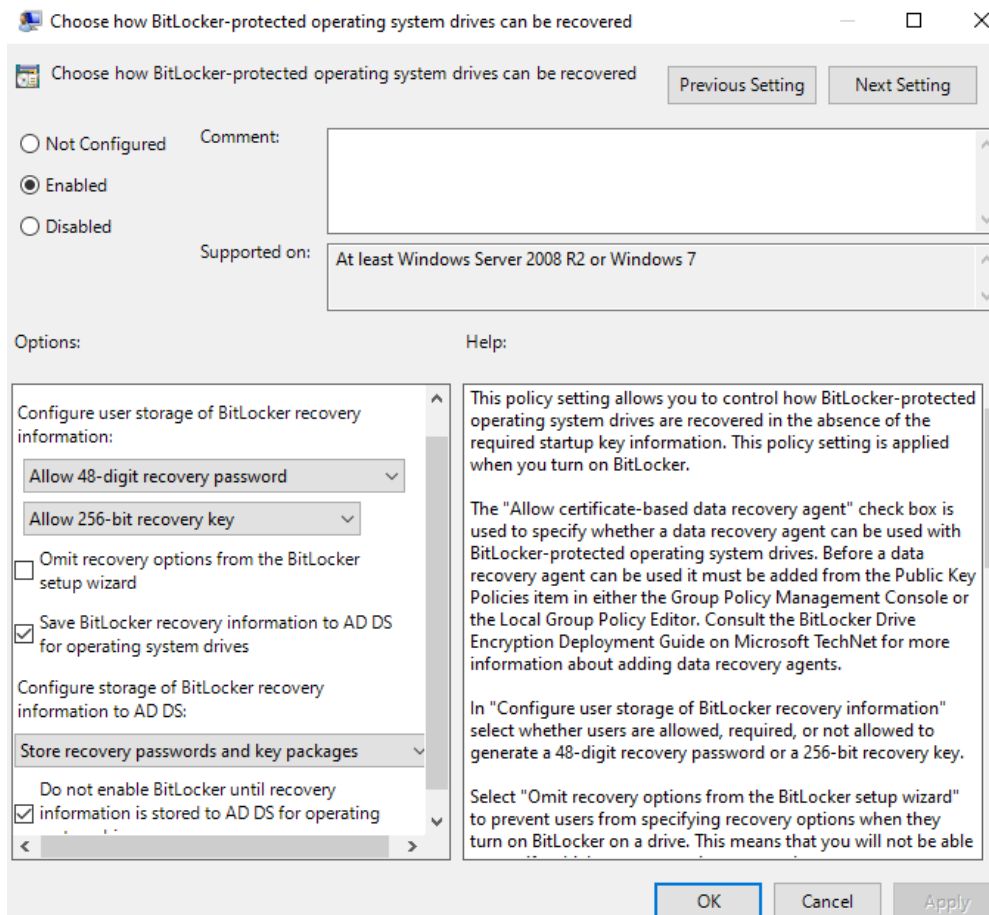
This policy is especially important because it applies specifically to the operating system drive, normally C:.

It allows the BitLocker recovery information for the system drive to be stored in Active Directory.

Policy path:

Computer Configuration → Policies → Administrative Templates → Windows Components → BitLocker Drive Encryption → Operating System Drives → Choose how BitLocker-protected operating system drives can be recovered





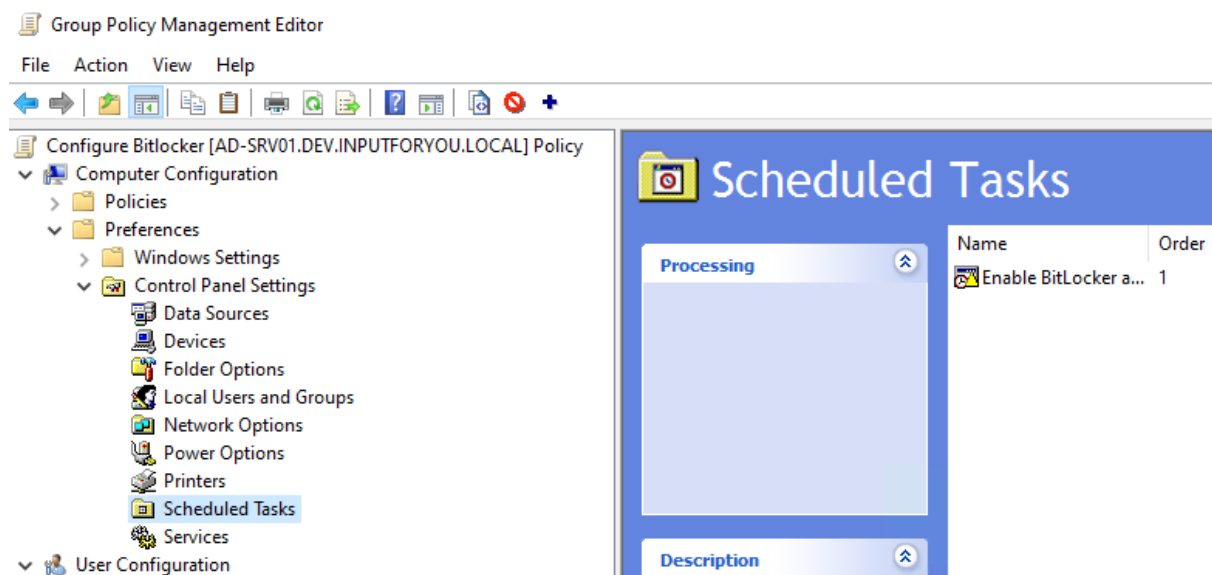
5. Scheduled task deployed by GPO

After configuring the BitLocker policies, a scheduled task is created through Group Policy.

This task runs the PowerShell script that backs up the BitLocker recovery key to Active Directory.

Task path in GPO:

Computer Configuration → Preferences → Control Panel Settings → Scheduled Tasks



Enable BitLocker and Backup Key to AD Properties

General Triggers Actions Conditions Settings Common

 Action: Update

Name: Enable BitLocker and Backup Key to AD

Author: HUGO\Administrator

Description: Automatically enables BitLocker and backs up the recovery key to Active Directory when network con

Security options

When running the task, use the following user account:
NT AUTHORITY\System Change User or Group...

☐ Run only when user is logged on

☒ Run whether user is logged on or not

☒ Do not store password. The task will only have access to local resources.

☐ Run with highest privileges

☒ Hidden

Configure for: Windows Vista™ or Windows Server™ 2008

OK Cancel Apply Help

New Trigger

Begin the task: On an event

Settings

☒ Basic

☐ Custom

Log: Microsoft-Windows-NetworkProfile/Operational

Source: Microsoft-Windows-NetworkProfile

Event ID: 10002

Advanced Settings

☒ Delay task for: 1 minute

☐ Repeat task every: 1 hour for a duration of: 1 day

☐ Stop all running tasks at end of repetition duration

☐ Stop task if it runs longer than: 3 days

☐ Activate: 11/05/2026 14:25:17 ☐ Synchronize across time zones

☐ Expire: 11/05/2026 14:25:17 ☐ Synchronize across time zones

☒ Enabled

OK Cancel

The path is : -NoProfile -ExecutionPolicy Bypass -File "\\WIN-OQFF2JIQ823.hugo.local\NETLOGON\BitLocker\Enable-BitLocker-And-Backup-To-AD.ps1"

The 'New Action' dialog box is shown with the title 'New Action' and a close button. It contains the instruction 'You must specify what action this task will perform.' Below this, the 'Action:' dropdown is set to 'Start a program'. Under the 'Settings' section, the 'Program/script:' field contains 'powershell.exe' with a 'Browse...' button next to it. The 'Add arguments(optional):' field contains '-NoProfile -ExecutionPolicy Bypass -File'. The 'Start in(optional):' field is empty. At the bottom are 'OK' and 'Cancel' buttons.

Enable BitLocker and Backup Key to AD Properties

The 'Settings' tab of the 'Enable BitLocker and Backup Key to AD Properties' dialog box is shown. It has tabs for 'General', 'Triggers', 'Actions', 'Conditions', 'Settings', and 'Common'. The 'Settings' section is titled 'Special additional settings that affect the behavior of the task.' and contains several options:

- ☒ Allow task to be run on demand
- ☐ Run task as soon as possible after a scheduled start is missed
- ☐ If the task fails, restart every: 15 minutes (dropdown). Below this, 'Attempt to restart up to: 3 times'.
- ☐ Stop the task if it runs longer than: 3 days (dropdown).
- ☐ If the running task does not end when requested, force it to stop
- ☐ If the task is not scheduled to run again, delete it after: 30 days (dropdown).

At the bottom, it says 'If the task is already running, then the following rule applies:' followed by a dropdown set to 'Do not start a new instance'. At the very bottom are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

The script it it on the path: \\filesrv01.dev.inputforyou.local\Software Distribution\BitLocker\Manually-Backup-BitLocker-RecoveryKey-To-AD.ps1

With this content:

```
<#
.SYNOPSIS
    Backs up the BitLocker recovery key to Active Directory.

.DESCRIPTION
    Finds the BitLocker RecoveryPassword protector on C: and backs it up to AD DS.
    BitLocker must already be enabled manually.
#>

$pcName = $env:COMPUTERNAME
$drive = "C:"
$logFile = "\\filesrv01.dev.inputforyou.local\Software Distribution\BitLocker\Logs\${pcName}.log"

function Write-Log {
    param([string]$message)

    $date = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
    "$date - $message" | Out-File -FilePath $logFile -Append -Encoding UTF8
}

try {
    $protectorOutput = manage-bde -protectors -get $drive -type RecoveryPassword 2>&1

    $protectorId = ($protectorOutput | Select-String "[0-9A-Fa-f-]{36}").Matches.Value | Select-Object
    -First 1

    if (-not $protectorId) {
        Write-Log "ERROR: No BitLocker recovery password found on $drive. BitLocker may not be
        enabled."
        exit 1
    }

    manage-bde -protectors -adbackup $drive -id $protectorId | Out-Null

    if ($LASTEXITCODE -eq 0) {
        Write-Log "SUCCESS: BitLocker recovery key backed up to AD. Protector ID: $protectorId"
        exit 0
    }
    else {
        Write-Log "ERROR: Failed to back up BitLocker recovery key to AD. Protector ID: $protectorId"
        exit 1
    }
}
catch {
    Write-Log "ERROR: $($_.Exception.Message)"
    exit 1
}
```